

A Framework for Unsupervised Planning of Cellular Networks Using Statistical Machine Learning

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Abstract—The wireless industry is moving towards developing smart cellular architectures that dynamically adjust the use of the network elements according to the service demand, and automating their operations in order to minimize both capital expenditure (CAPEX) and operation expenditure (OPEX). This involves developing efficient and unsupervised radio access network (RAN) planning, which has a direct impact on the system performance and CAPEX. This intelligent cellular planning aims at providing the base stations (BSs) configurations (e.g., coverage, user associations and antenna radiation pattern) that minimize the number of deployed BSs and meet the requirements in terms of coverage and capacity. The cellular planning optimization problem has been shown to be complex and non-scalable. Moreover, most of the existing cellular planning techniques result in an over or under provisioning architecture. Motivated by the above, we propose in this paper a novel and efficient unsupervised planning process. We make use of statistical machine learning (SML) to solve the problem at hand. The core idea of SML is that the planning parameters are treated as random variables. The parameters that maximize the corresponding joint probability distribution, conditioned on observations of users' positions, are learned or inferred using Gibbs sampling theory and Bayes' theory. To apply this theory to the planning problem, we make significant efforts to properly formulate the problem to be able to incorporate the constraints into the inference process and extract the planning parameters from the inferred model. Through several numerical examples, we compare the performance of the proposed approach to clustering-based and optimization-based existing planning approaches, and demonstrate the efficacy of our approach. We also demonstrate how our approach can leverage existing cellular infrastructures into the new design.

Index Terms—Bayesian inference, cellular network planning, self-organizing networks, statistical machine learning.

Manuscript received May 14, 2019; revised October 4, 2019 and December 30, 2019; accepted January 23, 2020. Date of publication February 5, 2020; date of current version May 15, 2020. This work was supported by Qatar National Research Fund (a member of Qatar Foundation) under NPRP Grant NPRP8-052-2-029, by an NSERC Grant, FQRNT Grant and Concordia University. The statements made herein are solely the responsibility of the authors. The associate editor coordinating the review of this article and approving it for publication was Z. Qin. (*Corresponding author: Mohaned Chraiti.*)

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Digital Object Identifier 10.1109/TCOMM.2020.2971691

I. INTRODUCTION

A. Motivation

THE wireless industry has been focusing on developing smart cellular architectures that dynamically adjust the use of the network elements according to the service demand, and automating their operations in order to minimize both capital expenditure (CAPEX) and operation expenditure (OPEX) [1]–[5]. A first step in this direction has already been taken by the Third-Generation Partnership Project (3GPP) and the Next Generation Mobile Networks Alliance (NGMN) in which they introduced the concept of Self-Organizing Network (SON), which is a technology designed to automate the planning, configuration, management, and healing operations in cellular networks. This concept received significant attention from the telecommunication leaders such as NOKIA and Ericsson. SON is anticipated to be worth 5.5 Billion by 2022 and to be the key enabling solution for low cost and high capacity 5G/6G networks [6].

An integral part of SON is the so-called radio access network (RAN) self-planning, which has received special attention owing to the fact that it considerably affects both the system performance and CAPEX [7], [8]. The need for self-planning is also aligned with the promotion of a dynamic RAN architecture that has the ability to adapt to the users' service demand. It is mainly enabled by the use of mobile BSs, such as unmanned aerial vehicle BSs (UAV-BSs) [9]–[11].¹ In this case, the task of planning could become time sensitive and costly, as it has to be performed rapidly and frequently. For example, we consider the scenario where mobile BSs (e.g., UAV-BSs) are used to support an existing terrestrial cellular network. To take advantage of the BSs mobility, their positions and parameters have to be judiciously updated according to the users' service demand. This implies that a considerable change on the users' distribution necessitates providing new deployment parameters. Traditional supervised planning techniques are based on human intervention and result in high OPEX. They could also be time consuming and therefore not suitable to use for rapid deployment.

Cell planning essentially involves identifying the network configurations (e.g., the BSs locations, and their antenna

¹In terms of standardization, a study on long-term evolution (LTE) network empowered by mobile BSs has been initiated by 3GPP in Release 15 [12]. In this study, LTE UAV-BS field test results are analyzed and documented for 6GHz bands [13]. Both academia and industry have demonstrated prototypes of UAV-BS capable cellular networks and made field test results [14].

radiation patterns) that minimizes the number of BSs subject to coverage, capacity and transmit power constraints. Additional facts have also to be considered including inter-cell interference. In practice, this is equivalent to providing a network setup that gives a good coverage while preventing resources over/under provisioning. In fact, a cellular system is said to be well planned (i.e., optimal plan) if it provides a good coverage, e.g., coverage per cell higher than 90%, and a good system utilization, e.g., utilization in each cell is between 70% and 90% of its maximum capacity [7], [8]. The lower bound on the cell utilization guarantees a good use of resources (i.e., prevents from resources over provisioning), whereas the other two constraints guarantee a good QoS (i.e., good coverage and non-saturated system).

B. Literature Review

The cellular planning problem is a classical problem, yet it leads to a non-scalable and complex optimization problem, that it has been shown to be NP-hard [15]. In fact, just identifying the locations of the BSs has been shown to be NP-Hard [15]. Nonetheless, several recent works aimed to develop unsupervised solutions that provide the essential planning parameters, namely, the BSs coverage, users clustering and the BSs radiation patterns (for the rest of the paper, we refer to those parameters simply as planning parameters), that minimize the number of the BSs to deploy while considering coverage, capacity and power constraints [16]–[27]. The key strategy adopted by these works consisted of decomposing the planning problem into sub-problems, by considering a sub-set of the aforementioned planning parameters and constraints, solving them separately, iteratively alternating between them and then manually removing/adding cells. For instance, the authors of [16]–[19] considered a fixed number of BSs to deploy, then they investigated their positions and/or users association. In [16], the authors used sphere packing to investigate the placement of multiple BSs to maximize the total coverage area for a given number of BSs. In [17], under the assumption of a priori known number of BSs and their locations, the authors used transport theory to cluster users, i.e., associate users to BSs. In [18] and [19], the authors considered the problem of deployment of one or two BSs to maximize the QoS.

In [20]–[23], the authors developed heuristic approaches to determine the number of BSs necessary to cover the entire area, i.e., coverage-based planning. Then, the architecture and/or users associations are updated according to other QoS criteria. In [21] and [22], the authors considered initially a large number of BSs. Then, they use swarm optimization to provide the BSs' positions considering the coverage constraint. Then, redundant BSs are removed, which leads to the final architecture. In [23], the authors divide the 3D space to be covered into octahedron shapes of equal dimensions. Each octahedron is considered as a cell. Now considering fixed BSs' positions, the authors used a kernel density estimation method (statistical learning approach) to build a statistical model on the users distribution. The inferred model is used to associate users to BSs in such a way that it minimizes the average latency. Along the same lines, the approach in [24] proposes to use the outage probability to determine the number of BSs and their locations. In [25], the planning

problem is divided into two sub-problems that are executed successively. First, considering a fixed number of BSs, their positions are optimized in such a way that the total used power is minimized. Second, the suggested algorithm identifies the redundant BSs and removes them (a BS is considered redundant if its absence does not cause constraints violation.) In [26], the key idea of the proposed approach is to first randomly place a large number of BSs, then to employ an iterative algorithm to eliminate the redundant BSs.

C. Problem Statement

Providing the essential planning parameters (i.e., configurations) that minimize the number of required BSs while satisfying the coverage, capacity and power constraints gives rise to a complex and non-scalable optimization problem (i.e., NP-hard) [15]–[22]. Existing related work clearly did not provide comprehensive and efficient solutions to the problem at hand where it was shown that the performance of those schemes fall short from reaching the optimal one, and in many cases, only sub-problems were addressed. For example, in [16]–[19], the goal was to maximize either the system capacity or coverage given certain limited resources. This naturally leads to resources being either over- or under-provisioned, i.e., the QoS requirements are not met or the number of BSs is much larger than the minimum. Moreover, those proposed solutions only solved sub-problems of the planning process while considering assumptions that may require human decisions, such as knowing the number BSs a priori. In [20]–[27], the main focus was to minimize the number of required BSs while satisfying the coverage requirement and providing enough resources to serve users. The network in this case could be over-provisioned (i.e., number of BSs much larger than the minimum required) given that there is no constraint on the minimum utilization per BSs. In light of the above, it is clear that none of the existing solutions is comprehensive enough to yield a configuration that minimizes number of required BSs while satisfying the cellular planning constraints, which is the focus of this paper.

D. Contributions

In this paper, we make several contributions that can be summarized in the following points.

- We develop an efficient and unsupervised cellular planning technique that yields a cellular network design (i.e., configuration), including the number of required BSs, their positions and their radiation antenna patterns, while satisfying constraints on capacity, coverage and power. It is shown through simulations that the number of required BSs is either equal to or larger than the optimal number by at most one.
- The proposed scheme invokes statistical machine learning (SML), namely, Bayesian learning, which requires to mathematically bind the probabilistic model of the users (i.e., observations) to latent variables such as the BSs positions and their radiation patterns, as well as to the constraints such as capacity, coverage and power [28]. Such adaptation involved significant derivations, and to the best of our knowledge, we are the first to provide such derivations.

- We show mathematically how the derived expressions can be exploited during the inference process to optimally update the model parameters and to help in the convergence to the resulting network configuration.
- We compare the performance of the proposed scheme to several existing schemes. We demonstrate through several examples that the proposed scheme reduces the number of required BSs considerably as compared to that of its counterpart schemes that exist in the literature.
- We also show that the proposed cellular planning scheme is flexible enough to be extended to existing static cellular infrastructures. In this case, during the planning process, the positions of the existing static BSs are fixed and additional mobile BSs are added to meet the network requirements while avoiding under- and over-provisioning of resources.

The rest of the paper is organized as follows. In Section II, we give a brief background on nonparametric Bayesian statistical learning. In Sections III and IV, the system model and the problem formulation are provided, respectively. The proposed algorithm is described in detail in Section V. We extend the proposed algorithm to the case where new BSs are deployed to support an existing architecture. The performance of the proposed approach is assessed and compared to existing benchmark approaches in Section VII. We conclude the paper in Section VIII.

II. BACKGROUND ON BAYESIAN STATISTICAL LEARNING

Machine learning approaches mainly fall in one of two categories, namely, data-driven (e.g., deep learning) and model-driven (e.g., Bayesian learning) approaches [28], [29]. Each of them has advantages over each other. The data-driven approaches optimize over a large training data set where mathematical models are not required. However, there is lack of control over convergence to the desired results, and it is challenging to incorporate constraints into the learning process. Meanwhile, model-driven approaches give the possibility to explicitly link the latent variables (that correspond for instance to the planning parameters of the problem at hand) as well as constraints on the model to be inferred from the available observations. Although such task could be challenging, incorporating these mathematical expressions in the inference algorithm helps in approaching the optimum solution.

Although there are many machine learning approaches existing in the literature, none of them have been exploited to solve the cellular planning problem considered in this paper. Moreover, the performance achieved by optimization-based cellular planning approaches is far from the optimal one. In this paper, we consider a model-driven approach, namely, Bayesian learning. This choice has been motivated by several reasons. First, Bayesian learning yields a structure that could be mathematically linked to the planning parameters and constraints [30]–[33], which suits the structure of the problem considered in this paper. This helps to explicitly incorporate the constraints into the inference process and hence converging to the intended results. Second, Bayesian learning offers the possibility to characterize probabilistically the gap to the

true posterior. Third, in terms of the performance achieved by Bayesian learning, we demonstrate in Sec. VII that this approach is able to approach the optimum planning setup. This suggests that even if we were to use another machine learning approach, such approach will be able, in the best case scenario, to achieve performance close to that of the proposed one.

In Bayesian statistics, all forms of uncertainty are expressed as randomness. Therefore, we model the unknown intended parameters, denoted by $\Theta = \{\theta_i, i \in \mathbb{N}\}$, as random variables, and they take values in space Ω . The model on observations $x_1, \dots, x_n$ suggests that they are generated in two stages. First, the parameters are sampled from Ω according to a prior known distribution G_0 . The prior gives the possibility to incorporate our thoughts, experiences, knowledge, etc, in how the parameters of the model should look like. For example, it is intuitive to conclude that BSs will most probably be located in areas where there is high density of users. Second, the data is independently sampled from the distribution P_Θ . That is,

$$\begin{aligned} \Theta &\sim G_0 \\ x_1, \dots, x_n | \Theta &\sim_{iid} P_\Theta. \end{aligned} \quad (1)$$

The objective now is to draw a conclusion about the values of Θ from the observations, which is provided through inferring the posterior distribution $G(\Theta) \triangleq P[\Theta | x_1, \dots, x_n]$ from which the most likely value of Θ is extracted. Using Bayes' rule, we have

$$G(\Theta \in \Omega) = \frac{\prod_{i=1}^n p(x_i | \Theta) G_0(\Theta)}{\int_{d\Theta \in \Omega} \prod_{i=1}^n p(x_i | d\Theta) G_0(d\Theta)}. \quad (2)$$

The explicit expression of the posterior is usually difficult to provide analytically. Nonetheless, to obtain the posterior and the values of the elements of Θ , there are inference approaches that can be used, such as Gibbs-sampling [30], [31], [33] (more on this in Sec. V.)

An inference model is said to be parametric if Ω has a finite dimension, denoted by K , that is known a priori. Obviously, this model is not suitable for our case since the number of parameters, which is essentially related to the number of required BSs, is unknown. In this case, Ω has to be of infinite dimensions and thus the inference model is said to be nonparametric.

III. SYSTEM MODEL

Consider a geographic area $\mathcal{D} \in \mathbb{R}^2$ in which a mobile network operator plans to provide wireless services for randomly distributed users. The operator aims at deploying BSs (B_k , $k \in \mathbb{N}$), that could be mobile, in the considered area to form a new cellular network fully based on new BSs. Later in this paper, we show how the proposed approach can be used to support a pre-deployed terrestrial cellular network.

A. Users Positions and Mobility

The planning parameters are inferred (i.e., learned) from a realization (sample, observation) of the users' positions at the time and area of interest. Such snapshot of the users' positions (x_{U_i}, y_{U_i}) could be obtained from existing statistics collected a priori by the service provider [7], [8]. This approach is basically similar to practical scenarios where the planning is

based on previously collected statistics. To elaborate, we consider the case of peak hours in different days where the traffic density is almost the same (some exceptions apply such as the case of special events, etc.) Then, a snapshot of the users' positions from a previous day could be used for planning in the following day for the same area and at the same time. To summarize, we assume that we have a snapshot of the approximate positions (x_{U_i}, y_{U_i}) of the N users $\{U_1, U_2, \dots, U_N\}$. This assumption is essentially used in almost all existing related work in the literature [16]–[22].

During the planning process, an upper bound on the utilization is usually set to 90% of the cell capacity. This guarantees that a slight increase in the wireless demand, up to 10%, can be accommodated by the cell without saturation. Moreover, different from existing planning techniques, the proposed scheme provides the planning parameters based on a probabilistic model that is built on the users distribution rather than the exact users' positions (as will be shown in Sec. V.) Thus, the network setup and performance are valid for any users' positions realizations sampled from the same density of users.

B. Channel Model and Interference

Although the proposed framework could be applied to any channel model, we need to choose a particular channel model and fix its parameters in order to describe in detail the different steps of the proposed solution and to provide the mathematical derivations (e.g., the outage probability) that are necessary to link the probabilistic model parameters to the planning parameters. In particular, we consider a path loss model that is widely used in the literature [7], [8], [20]–[22], where the path loss factor is equal to two. Without loss of generality, the path loss between user U_i and BS B_k can be written as

$$Los_{U_i, B_k} = \left(\frac{f}{(x_{U_i} - x_{B_k})^2 + (y_{U_i} - y_{B_k})^2} \right)^{-1}. \quad (3)$$

Here, f is a strictly positive real number that characterizes the effect of other factors such as the transmission frequency, shadowing, BS altitude, antennas gain, etc. For instance, if the free-space path loss model is adopted, f is equal to the square of the ratio of the wave length in meters over $4 \times \pi$ ($\pi \approx 3.14$) [7], [8], [20]–[22]. In this model, we also consider the widely used fading model, namely, Rayleigh fading [7], [8], [22].

The users in each cell are served in an orthogonal manner, i.e., there is no intra-cell interference. However, multiple cells may use the same frequency, giving rise to inter-cell interference, which is considered in our model. Obtaining exact expressions for the interference in downlink and uplink for each user may not be possible, given that it may require to track the instantaneous positions of users, which may not be possible. Nonetheless, in practice, multiple interference suppression and management techniques have been adopted to suppress or reduce the effect of inter-cell interference. Such techniques include frequency reuse techniques that are widely adopted in cellular systems whereby adjacent cells are prevented from transmitting over the same frequency. Therefore, the interference is usually caused only to/by users belonging to distant cells (i.e., non-adjacent cells). In this case, the distance

between a BS and an interfering user could be approximated by the distance between the BS of interest and the one that serves the concerned user. This leads to a simple and accurate approximation of the interference model that involves considering only the distance between BSs rather than the distance between each user and each BS. Such approximation has been adopted in related works in the literature [17]. Armed with the above argument, the interference at U_i associated to B_k can be approximated by

$$\begin{aligned} I_k &= \sum_{j=1, j \neq k}^K \varrho_{j,k} \frac{P_{B_j, B_k}}{Los_{B_j, B_k}} \\ &= \sum_{j=1, j \neq k}^K \varrho_{j,k} \frac{f \times P_{B_j, B_k}}{(x_{B_k} - x_{B_j})^2 + (y_{B_k} - y_{B_j})^2}, \end{aligned} \quad (4)$$

where P_{B_k, B_j} is the transmit power of B_j in the direction of a BS B_k . Here, $\varrho_{j,k}$ is used to characterize the effect of the deployed interference management technique, including frequency reuse techniques [7], interference alignment [34], interference dissolution [35], etc.

C. Performance Criterion: System Coverage and Capacity

One of the key performance measures that is considered in cellular planning is the outage probability. Another related performance is the admission probability, which is the complement of the outage probability. The admission probability per cell has to be, on average, higher than a threshold A_{th} for a given threshold on the received power, denoted by γ_{th} . The two thresholds depend on the service expected from the cellular system. In practice, a system is normally said to have a good coverage when $A_{th} \geq 90\%$ [7], [8].

As spectrum resources are limited, each BS can communicate simultaneously only with a limited number of users. Therefore, an upper bound on the number of users per cell has to be considered. Based on the available resources, the users traffic portfolio and random access model, one can provide the maximum number of users, denoted by $N_{\max}$, that can be supported by a BS. To guarantee that the system will accommodate a slight change in traffic and avoid system saturation, the number of users in a given cell has to be less than $c_{\max} N_{\max}$ [7], [8].

To avoid resources over-provisioning, a constraint on the minimum number of users per cell has to be considered. Usually a lower bound, defined as $c_{\min} N_{\max}$, is considered [7], [8]. A practical value of $c_{\min}$ is 70% [7], [8]. The lower bound is also used to characterize the gap to the optimal solution (the minimum number of required BSs). For instance, let us consider the scenario where there are N_T users to serve. Since a BS can serve up to $c_{\max} N_{\max}$ user, the number of required BSs should be higher than or equal to $\lceil \frac{N_T}{c_{\max} N_{\max}} \rceil$. Moreover, since the minimum number of users per cell is $c_{\min} N_{\max}$, the maximum number of BSs should be $\lceil \frac{N_T}{c_{\min} N_{\max}} \rceil$. As such, the gap between the maximum and the minimum number of BSs is $\lceil \frac{N_T}{c_{\min} N_{\max}} \rceil - \lceil \frac{N_T}{c_{\max} N_{\max}} \rceil$. Satisfying the above coverage and capacity constraints guarantees that the system provides a good coverage and good system usage, i.e., the system is well planned. These metrics are adopted as the main

performance evaluation criteria and are incorporated into the inference process.

IV. PROBLEM FORMULATION

The primary objective of our work is to provide the planning setup that minimizes the number of deployed BSs given certain coverage, power and capacity constraints. The planning parameters are inferred from a snapshot of the users' positions (observations). To do so, we adopt Bayesian learning as an intermediate stage in which the system infers (i.e., learn about) the parameters of the most accurate mathematical model for the users' distribution given the observations, while incorporating the considered constraints. Then, the planning parameters are extracted from the parameters of the inferred model. To achieve this goal, the parameters of the model have to be linked to the planning parameters as well as to the imposed constraints. Moreover, the inference model has to be carefully chosen. This is explained next.

A. Predictive Model: Dirichlet Process

In this paper, we use a Bayesian approach for inference [30], [33]. This requires defining a process from which the observations are sampled. In Bayesian statistical learning, the observations are assumed to be generated through a process that consists of two stages: first, the parameters Θ are sampled from certain distributions, then the observations are sampled from an obtained distribution that is defined through the sampled parameters. Now, we need to describe in detail the process from which the observations are sampled.

Defining a process includes defining the form of the predictive distribution of the users $P_{\Theta}(x, y)$, i.e., the shape of the users distribution. The users are randomly distributed and their density usually differs from a sub-area to another. Examining the histogram of the user positions may reveal peaks with different heights and widths, resembling a mixture of bivariate (2D) Gaussian, which can be used as an approximate shape. In this case, we have two sets of parameters: the parameters vector of the mixture elements $\theta = \{\theta_1, \theta_2, \dots\}$ and their weights $\pi = \{\pi_1, \pi_2, \dots\}$, i.e., $\Theta = \{\pi, \theta\}$. A bivariate Gaussian mixture has the following form.

$$P_{\Theta}(x, y) = \sum_k \pi_k \mathcal{N}_{\theta_k}(x, y), \quad (5)$$

where $\mathcal{N}_{\theta_k}(x, y)$ is the probability density function (PDF) of the bivariate Gaussian distribution given θ_k , which consists of the mean (x_{B_k}, y_{B_k}) and the covariance matrix $\text{COV}_{B_k} = \begin{bmatrix} \sigma_{k,x}^2 & \tau_k \sigma_{k,x} \sigma_{k,y} \\ \tau_k \sigma_{k,x} \sigma_{k,y} & \sigma_{k,y}^2 \end{bmatrix}$, $\{\tau_k \in [0, 1), \sigma_{k,x}, \sigma_{k,y} > 0\}$. That is,

$$\begin{aligned} \mathcal{N}_{\theta_k}(x, y) &= \frac{1}{2\pi \sigma_{k,x} \sigma_{k,y} \sqrt{1 - \tau_k^2}} \exp \left(-\frac{1}{2(1 - \tau_k^2)} \right. \\ &\quad \times \left. \left[\frac{(x_{B_k} - x)^2}{\sigma_{k,x}^2} + \frac{(y_{B_k} - y)^2}{\sigma_{k,y}^2} - \frac{2\tau_k(x_{B_k} - x)(y_{B_k} - y)}{\sigma_{k,x} \sigma_{k,y}} \right] \right). \end{aligned} \quad (6)$$

We note that the parameters to infer in (5) can be analytically associated to (i.e., computed from) those intended in the planning process as will be shown in Sec. IV-B.

Since we have two sets of parameters $\{\theta, \pi\}$, we have two parameter spaces $\theta \in \Omega_{\theta}$ and $\pi \in \Omega_{\pi}$, i.e., $\Omega = \{\Omega_{\theta}, \Omega_{\pi}\}$. We have $\Omega_{\pi} = \{[0, 1]^K | K \in \mathbb{N}, \sum_{k=1}^K \pi_k = 1\}$ [30], [33]. The elements of θ_k have the following form: $\{(x_{B_k}, y_{B_k}) \in \mathbb{R}^2, \tau_k \in (-1, 1), \sigma_{k,x}, \sigma_{k,y} > 0\}$. Define $\phi \triangleq \{\phi_i = \theta_k \text{ if } (x_{U_i}, y_{U_i}) \in k\text{th cluster}, i = [1, N_T]\}$ to be the latent vector of variables that is needed to associate each user to a cluster.

Let us now assume that we have a prior distribution G_0 for the elements of θ (explicit expression is provided in Sec. IV-C). Since we consider a mixture of distributions, the users' positions can be generated through a Dirichlet process, denoted by $DP(\alpha, G_0)$ (the proof can be found in [30], [33].) Here, α is the process precision which is real and strictly greater than zero. The process is defined as follows.

$$\begin{aligned} \pi_1, \pi_2, \dots &\sim \mathcal{D}(\alpha) \\ \theta_1, \theta_2, \dots &\sim G_0 \\ \phi_1, \phi_2, \dots, \phi_N | \theta, \pi &\sim \sum_k \pi_k \delta_{\theta_k} \\ (x_{U_i}, y_{U_i}) | \phi &\sim \mathcal{N}_{\phi_i}, \end{aligned} \quad (7)$$

where δ_{θ_k} is a Dirac measure and $\mathcal{D}(\alpha)$ is the Dirichlet distribution with parameter α . The choice of α will be discussed in Sec. V-A.

B. Model Parameters vs. Planning Parameters

We link in this section the model parameters (i.e., means and covariance matrices) and the planning parameters. Moreover, the effect of the capacity and coverage constraints on the parameters of the model is analyzed. This gives insight about the final intended values, which will help in the inference process described in Sec. V.

1) *BSs Number and Positions*: As mentioned earlier, the users can be clustered into K clusters based on the set of inferred parameters $\{\theta, \pi, \Phi\}$. Here, K is the length of the vector π (or θ). The K clusters can be seen as K cells. In this case, the number of required BSs is equal to K . Moreover, as the mean of each cluster is selected as the point that minimizes the outage probability, it is then judicious to associate the means to the BSs' 2D positions (x_{B_k}, y_{B_k}) .

2) *Outage Probability*: In the following, we first provide an exact expression for the average outage probability in each cell, which involves making the link between the parameters to be inferred and the BSs transmit power and antennas radiation patterns. Second, we transform the constraints γ_{th} and A_{th} into constraints on the elements of COV_{B_k} . To analyze the outage probability, we need to provide an expression for the received signal to interference plus noise ratio (SINR). That is,

$$\text{SINR}_{B_k, U_i} = \frac{f \times P_{B_k, U_i} |h_{B_k, U_i}|^2}{d_{B_k, U_i}^2} \frac{1}{\sigma_0^2 + I_k}, \quad (8)$$

where P_{B_k, U_i} is the transmit power of a BS B_k in the direction of a user U_i . h_{B_k, U_i} characterizes the multi-path effect which

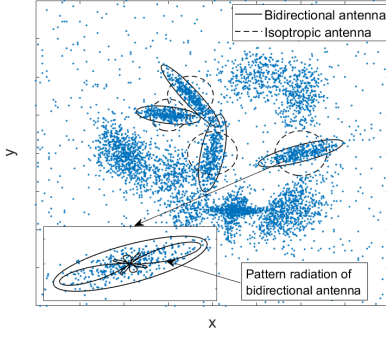


Fig. 1. BSs coverage shapes for different radiation patterns.

follows a Rayleigh distribution with a scale parameter that equals to one. $d_{B_k, U_i}^2 \stackrel{\delta}{=} (x_{B_k} - x_{U_i})^2 + (y_{B_k} - y_{U_i})^2$ is the $U_i - B_k$ Euclidean distance. The interference expression I_k is provided in (4). σ_0^2 characterizes the variance of the noise at the receiver.

Given some γ_{th} , the admission rate of U_i that is served by B_k is expressed as

$$\begin{aligned} A_{B_k}(x_{B_k}, y_{B_k}) &= 1 - P[\text{SINR}_{B_k, U_i} \leq \gamma_{th}] \\ &= 1 - P\left[|h_{B_k, U_i}|^2 \leq \gamma_{th}(\sigma_0^2 + I_k) \frac{d_{B_k, U_i}^2}{f \times P_{B_k, U_i}}\right] \\ &= \exp\left(-\gamma_{th}(\sigma_0^2 + I_k) \frac{d_{B_k, U_i}^2}{2f \times P_{B_k, U_i}}\right), \end{aligned} \quad (9)$$

where $\exp(\cdot)$ denotes the exponential function. From (9), the outage probability depends mainly on P_{B_k, U_i} , which may differ from one direction to another when directive antennas are used as in the case of our work. Now, we provide an explicit expression for P_{B_k, U_i} .

For the case when the BSs use isotropic antennas, the transmit power is similar in all directions. Thus, the factor $\frac{\gamma_{th}(\sigma_0^2 + I_k)}{f \times P_{B_k, U_i}}$ is similar for all users belonging to B_k , implying that the main factor that impacts the outage probability is the Euclidean distance between U_i and B_k . This suggests that a BS coverage (i.e., users cluster) will most likely take circular shape as shown in Fig. 1.

Employing directional antennas on BSs is beneficial as it may result in less interference and better coverage. In this paper, we consider bidirectional antennas where the radiation patterns form is depicted in Fig. 1.² In this case, the coverage of a BS will approximately take an elliptical shape, giving more flexibility compared to the case of isotropic antennas (see Fig. 1).

Since the transmit power may differ from one direction to another, we need to provide an expression for the BSs transmit power in all directions, which can be intractable, considering the exact form of the antenna radiation pattern. Moreover,

²A bidirectional antenna radiation pattern is considered in this paper, and this has been motivated by the fact that it has similarity with the base of the considered distribution of users, namely, the Gaussian distribution. It is also possible to consider other types of antennas by considering other distributions with bases similar to the intended antenna pattern shapes.

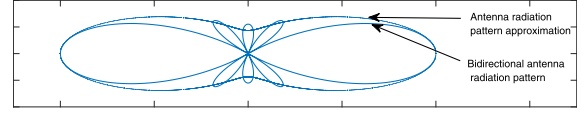


Fig. 2. Bidirectional antenna radiation pattern approximation.

characterizing such shape involves several variables, making it difficult to link to our model. As an alternative, we resort to approximating the radiation pattern of bidirectional antennas, as shown in Fig. 2. Such shape is obtained when the standard deviation forms an ellipse centred at the transmitter. Each point on the ellipse characterizes the standard deviation in the direction going from the center to the considered point. An ellipse can be well defined if its center, its axes lengths and their directions are provided. Hence, considering the above approximation, one can derive the exact expression of the transmit power in any direction. Mathematically, an ellipse can be defined by its center and a positive definite matrix [36]. In fact, let us consider an ellipse centred at (x_{B_k}, y_{B_k}) . The standard deviation can be defined by the following set of points.

$$\mathcal{S}_k = \{x, y \in \mathbb{R}^2 \mid [x_{B_k} - x \ y_{B_k} - y] \Sigma_{B_k}^{-1} \times [x_{B_k} - x \ y_{B_k} - y]^T = 1\}, \quad (10)$$

where $\Sigma_{B_k} = \begin{bmatrix} \zeta_{k,x}^2 & \tau'_k \zeta_{k,x} \zeta_{k,y} \\ \tau'_k \zeta_{k,x} \zeta_{k,y} & \zeta_{k,y}^2 \end{bmatrix}$ is a positive definite matrix. The parameters $\{\zeta_{k,x}, \zeta_{k,y}\}$ take values in $\mathbb{R}^2$, whereas $\tau'_k \in (0, 1)$. In this case, the ellipse axes directions and lengths are, respectively, the eigenvectors and the square root of the eigenvalues of Σ_{B_k} .

For a given standard deviation ellipse, P_{B_k, U_i} is expressed as follows.

Lemma 1: Considering that the standard deviation of the transmit power of B_k forms an ellipse defined by $\mathcal{S}_k$ in (10), we may express P_{B_k, U_i} as

$$P_{B_k, U_i} = \frac{(1 - (\tau'_k)^2) d_{B_k, U_i}^2}{\frac{(x_{B_k} - x_{U_i})^2}{\zeta_{k,x}^2} + \frac{(y_{B_k} - y_{U_i})^2}{\zeta_{k,y}^2} - \frac{2\tau'_k(x_{B_k} - x_{U_i})(y_{B_k} - y_{U_i})}{\zeta_{k,x} \zeta_{k,y}}}. \quad (11)$$

Proof: See Appendix A. $\square$

The transmit power from B_k to B_i can also be derived from Lemma 1 by replacing in (11) the coordinates of U_i by that of B_i , i.e., $(x_{U_i}, y_{U_i}) \rightarrow (x_{B_i}, y_{B_i})$. Then, one can provide an exact expression for the interference I_k in each cluster. Incorporating the exact expression of the transmit power in (9) gives an analytical expression for the admission rate per user as follows.

$$\begin{aligned} A_{B_k}(x_{U_i}, y_{U_i}) &= 1 - \exp\left(-\gamma_{th}(\sigma_0^2 + I_k) \frac{d_{B_k, U_i}^2}{2f \times P_{B_k, U_i}}\right) \\ &= \exp\left(-\frac{\gamma_{th}(\sigma_0^2 + I_k)}{2f \times (1 - (\tau'_k)^2)} \left[\frac{(x_{B_k} - x_{U_i})^2}{\zeta_{k,x}^2} + \frac{(y_{B_k} - y_{U_i})^2}{\zeta_{k,y}^2} - \frac{2\tau'_k(x_{B_k} - x_{U_i})(y_{B_k} - y_{U_i})}{\zeta_{k,x} \zeta_{k,y}} \right]\right). \end{aligned} \quad (12)$$

Each element (i.e., cluster) of the Gaussian mixture in (5) is a bivariate Gaussian. Then, the base of each element forms an ellipse for any given confidence interval [36]. The ellipse is centred at the mean (BS position), and the matrix that defines its base is proportional to its covariance matrix. Recall that, due to using bidirectional antenna, the standard deviation is characterized by an ellipse. The axes of this ellipse are parallel to and proportional to those of the cluster (see Fig. 1). Then, the matrix Σ_{B_k} is proportional to that of the cluster, which is in turn proportional to COV_{B_k} . Therefore, there exists a positive real number ξ_k such that $\Sigma_{B_k} = \xi_k \text{COV}_{B_k}$. This suggests that $\tau'_k = \tau_k$, $\zeta_{k,x}^2 = \xi_k \sigma_{k,x}^2$ and $\zeta_{k,y}^2 = \xi_k \sigma_{k,y}^2$. We use the results obtained above to derive an expression for A_{B_k} , as follows.

Lemma 2: The average admission rate in the k th cell is

$$A_{B_k} = \frac{1}{1 + \frac{\gamma_{th}(\sigma_0^2 + I_k)}{f \times \xi_k}}. \quad (13)$$

Proof: See Appendix B. $\square$

Recall that one objective is to have $A_{B_k} \geq A_{th}$ in each cell. We use this constraint to draw a lower bound on ξ_k as follows.

$$\begin{aligned} A_{B_k} \geq A_{th} &\Leftrightarrow \frac{1}{1 + \frac{\gamma_{th}(\sigma_0^2 + I_k)}{f \times \xi_k}} \geq A_{th} \\ &\Leftrightarrow \xi_k \geq \frac{A_{th} \gamma_{th} (I_k + \sigma_0^2)}{f \times (1 - A_{th})}. \end{aligned} \quad (14)$$

The expression in (14) defines a relation between Σ_{B_k} and COV_{B_k} in order to satisfy the coverage constraint. For a given COV_{B_k} , one can derive the minimum transmit power in each direction to achieve $A_{B_k} \geq A_{th}$. That is,

$$\zeta_{k,x}^2 \geq \frac{A_{th} \gamma_{th} (N_0 + I_k)}{f \times (1 - A_{th})} \sigma_{k,x}^2. \quad (15a)$$

$$\zeta_{k,y}^2 \geq \frac{A_{th} \gamma_{th} (\sigma_0^2 + I_k)}{f \times (1 - A_{th})} \sigma_{k,y}^2. \quad (15b)$$

Remark 1: Inequalities (15a) and (15b) play two major roles. First, they give the required transmission power and the intended radiation patterns once the parameters in COV_{B_k} are inferred. In fact, one can simply set $\zeta_{k,x}^2 = \frac{A_{th} \gamma_{th} (\sigma_0^2 + I_k)}{f \times (1 - A_{th})} \sigma_{k,x}^2$ and $\zeta_{k,y}^2 = \frac{A_{th} \gamma_{th} (\sigma_0^2 + I_k)}{f \times (1 - A_{th})} \sigma_{k,y}^2$ and this will guarantee that the outage probability constraint is satisfied. Moreover, τ_k will be deduced from COV_{B_k} . Second, such constraints are used to set an upper bound on $\{\sigma_{k,x}^2, \sigma_{k,y}^2, \tau_k\}$ in the case where there is a constraint on the BS transmit power as will be discussed next.

3) *Transmit Power Constraint:* We assume that there is a constraint on the overall transmit power standard deviation and it is given by $\pi \times P_{\max}$ ($\pi \simeq 3.14$), i.e., the total standard deviation of an isotropic antenna, where the standard deviation in each direction is equal to $\sqrt{P_{\max}}$. This constraint has to be linked to the parameters to be inferred. Assuming that B_k transmits in an isotropic fashion, then the overall standard deviation is the surface of a circle with radius $\sqrt{P_{B_k, x U_i}}$, i.e., $\pi \times P_{B_k, x U_i}$. In this case, P_{B_k, U_i} has to be less than or equal to $P_{\max}$.

The standard deviation of the transmit power takes an elliptical shape with axes lengths equal to $2\sqrt{\lambda_{1,k}}$ and $2\sqrt{\lambda_{2,k}}$,

where $\lambda_{1,k}$ and $\lambda_{2,k}$ are the eigenvalues of Σ_{B_k} whose expressions are provided in (33) and (34) in Appendix A. In this case, the overall transmit power can be characterized through the surface of the ellipse. Considering the constraint on the power $\pi \times P_{\max}$, we have

$$\begin{aligned} \pi \times \sqrt{\lambda_{1,k} \lambda_{2,k}} &\leq \pi \times P_{\max} \\ &\Leftrightarrow \frac{\zeta_{k,x}^2 + \zeta_{k,y}^2 + \sqrt{(\zeta_{k,x}^2 + \zeta_{k,y}^2)^2 - 4(1 - \tau_k^2) \zeta_{k,x}^2 \zeta_{k,y}^2}}{2} \\ &\quad \times \frac{\zeta_{k,x}^2 + \zeta_{k,y}^2 - \sqrt{(\zeta_{k,x}^2 + \zeta_{k,y}^2)^2 - 4(1 - \tau_k^2) \zeta_{k,x}^2 \zeta_{k,y}^2}}{2} \\ &\leq P_{\max}^2 \\ &\Leftrightarrow \frac{(\zeta_{k,x}^2 + \zeta_{k,y}^2)^2 - \left((\zeta_{k,x}^2 + \zeta_{k,y}^2)^2 - 4(1 - \tau_k^2) \zeta_{k,x}^2 \zeta_{k,y}^2 \right)}{4} \\ &\leq P_{\max}^2 \\ &\Leftrightarrow (1 - \tau_k^2) \zeta_{k,x}^2 \zeta_{k,y}^2 \leq P_{\max}^2. \end{aligned} \quad (16)$$

Combining the expressions in (15) and (16), we can obtain an upper bound on COV_{B_k} as follows.

$$\begin{aligned} (1 - \tau_k^2) \sigma_{k,x}^2 \sigma_{k,y}^2 \left(\frac{A_{th} \gamma_{th} (\sigma_0^2 + I_k)}{f \times (1 - A_{th})} \right)^2 &\leq P_{\max}^2 \\ \Rightarrow (1 - \tau_k^2) \sigma_{k,x}^2 \sigma_{k,y}^2 &\leq P_{\max}^2 \left(\frac{f \times (1 - A_{th})}{A_{th} \gamma_{th} (\sigma_0^2 + I_k)} \right)^2. \end{aligned} \quad (17)$$

The condition in (17) is sufficient to guarantee that the coverage constraint is satisfied while respecting the transmit power constraint.

4) *On the Cell Capacity Constraints:* In order to avoid cell congestion as well as resources over provisioning, we set lower and upper bounds on utilization per cell, i.e., $c_{\min} N_{\max} \leq N_k \leq c_{\max} N_{\max}$, where N_k is the number of users associated to B_k . This suggests that the elements of the weight vector $\{\pi_1, \pi_2, \dots\}$ have to be in the following range $\frac{c_{\min} N_{\max}}{N_T} \leq \frac{N_k}{N_T} = \pi_k \leq \frac{c_{\max} N_{\max}}{N_T}$.

The cell size and the number of users associated to it normally scale with the transmit power. As described in Sec. IV-B.3, adjusting the transmit power is equivalent to adjusting the elements of the covariance matrix that characterizes the cell. We describe next how it is possible to tune the covariance matrix elements to approach the desired number of users for the k th cluster.

Let N_k be the number of users belonging to the k th cluster that is characterized by the covariance matrix Σ_{B_k} . If $r_{N_k, \max} = \frac{N_k}{c_{\max} N_{\max}} \geq 1$ or $r_{N_k, \min} = \frac{N_k}{c_{\min} N_{\max}} \leq 1$, then we have interest to respectively increase or decrease the cluster size while maintaining the same radiation pattern shape. As discussed in Sec. IV, the cluster takes an elliptical shape. Given the bivariate Gaussian distribution, the surface of the ellipse that contains a given percentage of users (i.e., a given confidence interval) is proportional to $\pi \sqrt{\lambda_{k,x} \lambda_{k,y}}$, where $\lambda_{k,x}$ and $\lambda_{k,y}$ are the eigenvalues of Σ_{B_k} . To approach the intended number of users per cluster, $r_{N_k, (\cdot)} N_k$, we can decrease or increase the surface covered by

the cluster by the factor $r_{N_k,(\cdot)}$. The new cluster coverage becomes, $r_{N_k,(\cdot)} \text{pi} \sqrt{\lambda_{k,x} \lambda_{k,x}} = \text{pi} \sqrt{r_{N_k,(\cdot)} \lambda_{k,x} r_{N_k,(\cdot)} \lambda_{k,x}}$. This is the surface of the ellipse with covariance matrix $r_{N_k,(\cdot)} \Sigma_{B_k}$.

There is another parameter that can affect the final number of BSs, namely, the process precision α . In fact, the final number of BSs scales with α . Therefore, this parameter has to be well tuned during the inference process (as described in detail in Sec. V).

C. The Prior

The last missing piece to completely define the Dirichlet process is the prior $G_0(\theta)$. Recall that the prior is the possible distribution over the means $\{(x_{B_k}, y_{B_k}) | k \in \mathbb{N}\}$ and the covariance matrices $\{\text{COV}_{B_k} | k \in \mathbb{N}\}$. Since the means and the covariance matrices are independent, $G_0(\theta)$ is simply the product of the individual distributions. Here, there are two major challenges to be addressed. First, we must define the appropriate priors, since arbitrarily choosing the prior will considerably contaminate the final distribution. Second, during the inference process (as will be discussed in Sec. V), we need to provide a close form expression for $p(x_{U_i}, y_{U_i})$ given G_0 , i.e.,

$$p(x_{U_i}, y_{U_i} | G_0) = \int_{\theta \in \Omega} p(x_{U_i}, y_{U_i} | \theta) G_0(\theta) d\theta, \quad (18)$$

for each user. Therefore, we have also interest in choosing the prior distribution such that the integral in (18) is tractable.

The BSs positions are likely to be located in areas where there are concentrations of users. Based on this observation, a legitimate choice of the prior distribution over the means is a mixture of Gaussians where the mixture weights are high in the areas with high users densities. Let us denote the number of mixture elements by K_0 , the means by $\mathbf{m}_0 = \{m_{0,1}, m_{0,2}, \dots, m_{0,K_0}\}$, and the covariance matrices by $\Lambda_0 = \{\Lambda_{0,1}, \Lambda_{0,2}, \dots, \Lambda_{0,K_0}\}$. We also use $\pi_0 = \{\pi_{0,1}, \pi_{0,2}, \dots, \pi_{0,K_0}\}$ to denote the mixture weights vector. That is,

$$(x_{B_k}, y_{B_k}) | \pi_0, \mathbf{m}_0, \Lambda_0 \sim \sum_{k=1}^{K_0} \frac{1}{\pi_{0,k}} \mathcal{N}_{m_{0,k}, \Lambda_{0,k}}(\cdot). \quad (19)$$

The parameters $\{\pi_0, \mathbf{m}_0, \Lambda_0\}$ are called hyper-parameters and they have to be known a priori. The parameters K_0 and $\mathbf{m}_0$ may be chosen from the histogram of the observations, where K_0 would be the number of peaks and $\mathbf{m}_0$ would be the 2D positions of those peaks. As for Λ_0 , it is difficult to obtain from the histogram. As such, to account for its uncertainty, we treat its elements as random matrices.

To make the integral in (18) tractable, we link the distributions of COV_{B_k} to that of Λ_0 . Next, we provide the distribution of COV_{B_k} . As shown in Sec. IV-B.2 and Sec. IV-B.3, the antenna radiation patterns are defined through the elements of COV_{B_k} , but they are unknown. Therefore, they can be approximated by the covariance matrix of an isotropic antenna defined by $\text{COV}_0 = P_{\max} I_{2 \times 2}$ with some uncertainty, where $I_{2 \times 2}$ is the 2×2 identity matrix. In this case, the distribution of

COV_{B_k} is the Wishart distribution with parameters COV_0 and of degree three $\mathcal{W}_{\text{COV}_0, 3}(\cdot)$ [37]. Indeed, the number of degrees being three comes from the fact that COV_{B_k} is symmetric and can be defined via three elements; $\sigma_{k,x}^2$, $\sigma_{k,y}^2$ and $(1 - \tau_k) \sigma_{k,x} \sigma_{k,y}^2$. That is,

$$\text{COV}_{B_k} \sim \mathcal{W}_{\text{COV}_0, 3}(\cdot) = \frac{\exp[-\text{tr}(\text{COV}_0^{-1} \times \cdot) / 2]}{2^3 |\text{COV}_0|^{\frac{3}{2}} \Gamma_2(3/2)}, \quad (20)$$

where $\text{tr}(\cdot)$ and $|\cdot|$ denote the trace and the determinant operators, respectively.

In addition to the advantage of giving a good prior for COV_{B_k} , the Wishart distribution is well known to be a conjugate for the Gaussian distribution, which should help in getting a closed form expression for the integral in (18). Let us assume that there exists a positive constant ϖ such that the elements of $\frac{1}{\varpi} \Lambda_0$ follow $\mathcal{W}(\text{COV}_0, 3)$, i.e., $\frac{1}{\varpi} \Lambda_{0,k} \sim \mathcal{W}_{\text{COV}_0, 3}$. In this case, we have

$$\begin{aligned} & G_0(x_{B_K}, y_{B_K}, \text{COV}_{B_K}) \\ &= \sum_{j=1}^{K_0} \frac{1}{\pi_{0,j}} \mathcal{N}_{m_{0,j}, \frac{1}{\varpi} \text{COV}_{B_K}}(x_{B_K}, y_{B_K} | \frac{1}{\varpi} \text{COV}_{B_K}) \\ & \quad \mathcal{W}_{\text{COV}_0, 3}(\text{COV}_{B_K}). \end{aligned} \quad (21)$$

Armed with the above results, the integral in (18) is viewed as a mixture of T-distributions [38]:

$$\begin{aligned} p(x_{U_i}, y_{U_i} | G_0) &= \int_{\Omega_{\theta}} p(x_{U_i}, y_{U_i} | x_{B_K}, y_{B_K}, \text{COV}_{B_K}) \\ & \quad \times G_0(x_{B_K}, y_{B_K}, \text{COV}_{B_K}) dx_{B_K}, y_{B_K}, \text{COV}_{B_K} \\ &= \sum_{j=1}^{K_0} \int_{\Omega_{\theta}} p(x_{U_i}, y_{U_i} | x_{B_K}, y_{B_K}, \text{COV}_{B_K}) \\ & \quad \times \frac{1}{\pi_{0,j}} \mathcal{N}_{m_{0,j}, \frac{1}{\varpi} \text{COV}_{B_K}}(x_{B_K}, y_{B_K} | \frac{1}{\varpi} \text{COV}_{B_K}) \\ & \quad \times \mathcal{W}_{\text{COV}_0, 3}(\text{COV}_{B_K}) dx_{B_K}, y_{B_K}, \text{COV}_{B_K} \\ &\stackrel{b}{=} \sum_{j=1}^{K_0} \frac{1}{\pi_{0,j}} \mathcal{T}_{m_{1,j}, t, \tau}(x_{U_i}, y_{U_i}), \end{aligned} \quad (22)$$

where $t = \frac{5\varpi}{2(1+\varpi)} \text{COV}_0^{-1}$. The equality (b) is obtained by invoking the results in [39].

V. INFERRING THE PLANNING PARAMETERS

The planning parameters can be computed from the true posterior $G(\phi_n) = \sum_{k \in \mathbb{N}} \pi_k \delta_{\theta_k}(\phi_n)$. However, the true posterior is difficult to compute analytically using Bayes' rule. Meanwhile, there are low inference algorithms that can provide the posterior such as the widely used Gibbs sampling [30], [33]. It has been shown that Gibbs sampling can be executed in linear time [40]. In the case of a Dirichlet process, the Gibbs sampling approach is based on the Ferguson theorem [30]–[32]. It states that following the Gibbs sampling the true posterior takes the form

$$G \sim \frac{\alpha}{\alpha + N} G_0 + \frac{1}{\alpha + N} \sum_{n=1}^N \delta_{\phi_n}. \quad (23)$$

The theorem suggests that Gibbs sampling converges to the true posterior as the number of observations increases and the gap is at most $\frac{\alpha}{\alpha+N}G_0$. Indeed, the second term in (23) is considered as the posterior and hence the gap to the true one corresponds to the second term, that is $\frac{\alpha}{\alpha+N}G_0$. Now, it becomes clear that the gap decreases with the number of users N and hence convergence to the true posterior.

We adopt in this paper, an algorithm based on the MacEachern's Gibbs sampling algorithm, which offers faster convergence compared to the naive Gibbs sampling algorithm [30]. In the MacEachern's algorithm, two steps are executed iteratively: associating each user to an existing cluster or generating a new one, and updating the parameters of each cluster. In each of these steps, we make use of our finding in Sec. IV. The MacEachern's algorithm gives results for a given process precision α . Along with the inference process, we adjust the value of α using the bisection algorithm until all the constraints are satisfied. The proposed algorithm outline is described in Algorithm 1. In the algorithm, N_{iter} and ϕ^{-i} denote the number of iterations and the vector that contains the elements of ϕ except for the one associated to U_i (i.e., ϕ_i).

Algorithm 1 Proposed Algorithm Outline

Initialization: α, N_{iter} ;
while Constraints on coverage and capacity are not satisfied **do**
 Update α ;
 MacEachern Algorithm:
 for $L = 1 \rightarrow N_{iter}$ **do**
 for $i = 1 \rightarrow N$ **do**
 $P[\phi_i | \phi^{-i}, x_{U_i}, y_{U_i}] =$
 $\begin{cases} \frac{\alpha}{N + \alpha} \int_{\phi_i \in \Omega_\theta} P[x_{U_i}, y_{U_i} | \phi_i] G_0(\phi_i) d\phi_i, & \phi_i \leftarrow \theta_{new} \\ \frac{\sum_{j=1, j \neq i}^N \delta_{\theta_k}(\phi_j)}{N + \alpha} P[x_{U_i}, y_{U_i} | \theta_k] \forall \theta_k \in \theta, & \phi_i \leftarrow \theta_k \end{cases}$
 $\phi_i \leftarrow \arg \max_{\phi_i \in \Omega_\theta} P[\phi_i | \phi^{-i}, \theta, \pi, x_{U_i}, y_{U_i}]$
 if $Boolean(\phi_i \leftarrow \theta_{new}) \leftarrow True$ **then**
 $K \leftarrow K + 1$
 $\theta \leftarrow \{\theta, \theta_{new}\}$
 for $k = 1 \rightarrow K$ **do**
 Update the parameter of k th cluster:
 $(x_{B_k}, y_{B_k}, COV_{B_k})$
 Performance Analysis: Performance analysis in terms of coverage and capacity

We discuss next in detail each step of the algorithm, where we show how the results obtained in Sec. IV are used in the algorithm.

A. Initialization of α

Considering a Dirichlet process and a number of observations N_T , the average number of generated clusters is equal to $\sum_{n=1}^{N_T} \frac{\alpha}{\alpha+n-1} \simeq \alpha \log\left(\frac{N_T}{\alpha}\right)$ [32]. Moreover, from the cell

capacity upper bound, we know that the minimum number of clusters is $\lceil \frac{N_T}{c_{\max} N_{\max}} \rceil$. Therefore, we choose α such that $\frac{N_T}{c_{\max} N_{\max}} = \alpha \log\left(\frac{N_T}{\alpha}\right)$. This gives

$$\alpha = -\frac{N_T}{c_{\max} N_{\max} L(-1/(c_{\max} N_{\max}))}, \quad (24)$$

where $L(\cdot)$ is the Lambert function [41]. As for θ , the vector of the mixture Gaussian parameters, reasonable initial values would be inspired from the histogram of the observations.

B. User Association

During the user association step, a user is either associated to one of the existing clusters or to a new one. For each user U_i , we compute $P[\phi_i | \phi^{-i}, x_{U_i}, y_{U_i}]$. A user is associated to an existing cluster B_k with probability

$$P[\phi_i = \theta_k | \phi^{-i}, x_{U_i}, y_{U_i}] = \frac{\sum_{j=1, j \neq i}^N \delta_{\theta_k}(\phi_j)}{N + \alpha} P[x_{U_i}, y_{U_i} | \theta_k], \quad (25)$$

where $P[x_{U_i}, y_{U_i} | \theta_k] \sim \mathcal{N}_{\theta_k}$, or to a new cluster with probability

$$\begin{aligned} &P[\phi_i = \theta_{new} | \phi^{-i}, x_{U_i}, y_{U_i}] \\ &= \frac{\alpha}{N + \alpha} \int_{\phi_i \in \Omega_\theta} P[x_{U_i}, y_{U_i} | \phi_i] G_0(\phi_i) d\phi_i \\ &\stackrel{c}{=} \frac{\alpha}{N + \alpha} \sum_{j=1}^{K_0} \frac{1}{\pi_{0,j}} \mathcal{T}_{m_1, j, t, 7}(x_{U_i}, y_{U_i}), \end{aligned} \quad (26)$$

where equality (c) comes from our derivation in (22) in Sec. IV-C (the T-distribution parameters are defined below (22)). Then, the value of ϕ_i with the maximum probability is selected. This is equivalent to associating U_i to the BS that minimizes its outage probability. In the case when $\phi_i \leftarrow \theta_{new}$, a new randomly generated cluster is created and the total number of clusters increases by one. The parameters of the new cluster are defined through θ_{new} , which consists of a randomly chosen mean $(x_{B_{new}}, y_{B_{new}})$ and covariance $COV_{B_{new}} = P_{\max} I_{2 \times 2}$.

C. Clustering Parameters Update

The mean of a cluster corresponds to the point that minimizes the users outage probability:

$$(x_{B_k}, y_{B_k}) = \frac{1}{\sum_{i=1}^N \delta_{\theta_k}(\phi_i)} \sum_{i=1}^N (x_{U_i}, y_{U_i}) \delta_{\theta_k}(\phi_i). \quad (27)$$

For the covariance matrices, they are refined through multiple stages. In the first stage, the algorithm computes the most likely covariance matrix COV_{B_k} given the data. That is,

$$\sigma_{k,x}^2 = \frac{\sum_{i=1}^N (x_{U_i} - x_{B_k})^2 \delta_{\theta_k}(\phi_i)}{\sum_{i=1}^N \delta_{\theta_k}(\phi_i)} \quad (28a)$$

$$\sigma_{k,y}^2 = \frac{\sum_{i=1}^N (y_{U_i} - y_{B_k})^2 \delta_{\theta_k}(\phi_i)}{\sum_{i=1}^N \delta_{\theta_k}(\phi_i)} \quad (28b)$$

$$\tau_k \sigma_{k,x} \sigma_{k,y} = \frac{\sum_{i=1}^N (x_{U_i} - x_{B_k})(y_{U_i} - y_{B_k}) \delta_{\theta_k}(\phi_i)}{\sum_{i=1}^N \delta_{\theta_k}(\phi_i)}. \quad (28c)$$

We then adjust the parameters of the covariance matrices to approach the intended constraints on the capacity per cell. As discussed in Sec. IV-B.4, if the number of elements N_k is far from the bounds by factor $r_{N_k,(\cdot)}$, we update the covariance matrix as $\text{COV}_{B_k} \leftarrow r_{N_k,(\cdot)} \text{COV}_{B_k}$. Considering the last updated values of COV_{B_k} , we verify the joint constraint on the coverage and the transmit power provided in (17). Recall that (17) means that it is possible to guarantee that the average coverage per cell is higher than or equal to A_{th} while respecting the power constraint. In the case when the power constraint is not satisfied, we consider the most likely distribution $\mathcal{N}_{\theta_k}$ that would satisfy the power constraint. To this end, we derive the normal distribution $\mathcal{N}_{\theta'_k}$ that minimizes the Kullback-Leibler divergence distance to $\mathcal{N}_{\theta_k}$ while respecting the power constraint [42].

Lemma 3: Given a normal distribution $\mathcal{N}_{\theta_k}$ with a covariance matrix that does not verify the constraint in (17), the covariance matrix $\text{COV}'_{B_k} = \begin{bmatrix} \sigma'_{k,x} & \tau'_k \sigma'_{k,x} \sigma'_{k,y} \\ \tau'_k \sigma'_{k,x} \sigma'_{k,y} & \sigma'_{k,y} \end{bmatrix}$ of $\mathcal{N}_{\theta'_k}$ that minimizes the Kullback-Leibler divergence given the constraints in (17) is defined as follows.

$$(\tau'_k = \tau_k, \quad \sigma'_{k,x} = \mu \sigma_{k,x}, \quad \sigma'_{k,y} = \mu \sigma_{k,y}), \quad (29)$$

$$\text{where } \mu = \frac{P_{\max}}{\sqrt{1 - \tau_k^2 \sigma_{k,x} \sigma_{k,y}}} \left(\frac{f \times (1 - A_{th})}{A_{th} \gamma_{th} (\sigma_0^2 + I_k)} \right).$$

Proof: See Appendix C. $\square$

To summarize, the parameters of each cluster are first updated based on the positions of the users belonging to that cluster. Then, the constraint on the capacity per cell is verified. Otherwise, the covariance matrix is updated as $\text{COV}_{B_k} \leftarrow r_{N_k,(\cdot)} \text{COV}_{B_k}$.

D. Fine Tuning of α

For each α , at the end of the algorithm, if the capacity and coverage per cell are satisfied in all cells, the while loop ends. Otherwise, the value of α increases or decreases depending on the obtained results. If the system is saturated and/or the coverage constraint is not satisfied, α increases. In the case when the resources are over provisioned, α decreases. We make use of the bisection algorithm to update α [43].

It may happen that the system is coverage limited and it is not possible to satisfy the lower bound on the cell capacity. Such cell will be exempted from the capacity constraint. The question is how one can know that a cell is coverage limited. This can be done by analyzing its transmit power. If it uses the maximum transmit power, i.e., $(1 - \tau_k^2) \zeta_{k,x}^2 \zeta_{k,y}^2 = P_{\max}^2$, and the number of users remains lower than the minimum required, it is considered as a coverage limited cell.

VI. SUPPORTING EXISTING CELLULAR SYSTEMS

Throughout the development of our framework, we have considered the scenario in which the cellular network design is intended for a new area (i.e., clean slate). However, as mentioned before, the proposed framework works as well for existing cellular infrastructures whereby new BSs may need to be deployed permanently or temporarily. The latter scenario

may be encountered in certain areas during special events, peak hours, BS failure, etc. The objective therefore becomes to augment an existing configuration with additional BSs.

We assume that there are K_e existing BSs $\{B_1, B_2, \dots, B_{K_e}\}$. As a first step, we associate the users to one of the existing cells. Then, we compute the bivariate distribution that represents the user in each cell, where the distribution means are the terrestrial BSs positions. The covariance matrices parameters can be found as described in (28). Then, considering the coverage and constraint parameters, the elements of COV_{B_k} are updated as described in Lemma 3. Once we obtain the updated distribution parameters, we incorporate this knowledge into the prior. In fact, rather than starting with prior G_0 , we consider the following prior.

$$G'_0 = \frac{\alpha}{N_T + \alpha} G_0 + \frac{1}{N_T + \alpha} \sum_{k=1}^{K_e} N_k \delta_{\theta_k}. \quad (30)$$

This has the form of a posterior of a Dirichlet process given prior G_0 (see (23).) As per the Ferguson theorem [30]–[32], the posterior also follows a Dirichlet process. Then, we proceed exactly as described in Algorithm 1. The only main difference is that we do not update the means of the cluster $\{B_1, B_2, \dots, B_{K_e}\}$. The final posterior will be of the form.

VII. SIMULATION RESULTS AND DISCUSSION

In this section, we assess the performance of the proposed SML planning technique and we compare its performance to that of existing related work, including machine learning-based and optimization-based techniques. As for the optimization-based techniques, we consider the approach proposed in [21], [22] where the authors considered objectives and constraints similar to the ones in this paper. Specifically, the technique is based on Swarm intelligence, where it initially randomly generates a large number of possible plans, called particles, while considering a large number of BSs. For multiple iterations, they repeat the following process. Each particle is analyzed in terms of coverage and capacity. The particle with the best configuration (i.e., BSs positions) is kept and used to update the other particles' configurations. At the end of the process, they apply an elimination algorithm that removes redundant BSs which leads to the final architecture.

As for the existing machine learning-based planning approaches, their key idea is to cluster (i.e., label) the users and then the obtained clusters are considered as cells. We remark that there are many machine learning techniques, however, it is not clear how to adopt any of them to solve appropriately the cellular planning problem considered here and, as mentioned above, this may require substantial adaptation, similar to what was done in this paper for the SML approach. In light of this, we select two non-parametric learning (i.e., clustering) techniques where only straightforward derivations are considered as described in the following. The first one is based on the K-mean clustering approach that is widely adopted in cellular planning. We consider the K-mean approach that

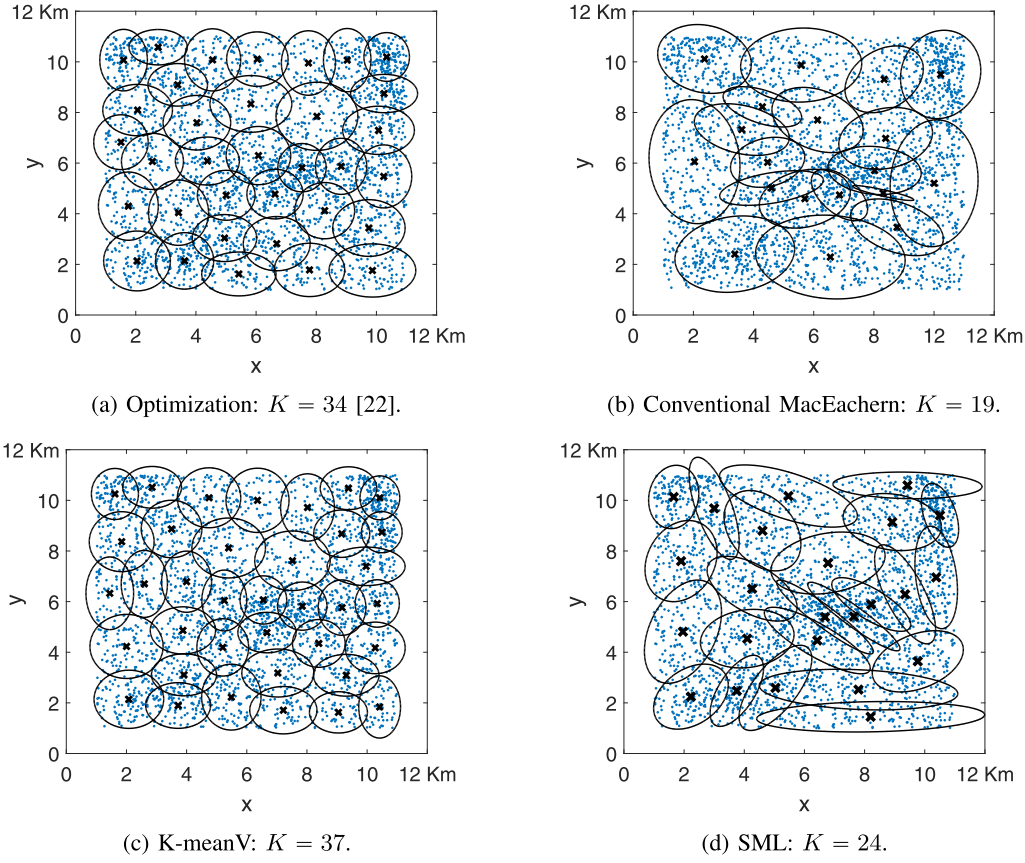


Fig. 3. Dense urban area: users distribution and association.

usually requires fixing the number of clusters, which is typically unknown. To solve this, we consider K-mean with a varying number of BSs (denoted here by K-meanV) in which it starts by a low number of BSs and then it keeps increasing the number as long as the coverage per cell and/or the upper bound constraint on the maximum number of users per cell are not satisfied. This will guarantee that the entire area will be covered and the users get enough resources. The second technique is based on the MacEachern algorithm, which is a non-parametric learning technique. In fact, we consider the conventional MacEachern algorithm without adapting it to incorporate the constraints on the capacity and coverage. That is, we consider only the straightforward link in which we consider that the clusters and their means correspond to the BSs and their positions, respectively. This is done to highlight the importance of the different derivations and contributions that are made in this paper to obtain a good planning framework. We consider the same prior adopted in the paper and the same parameters (e.g., the process precision α) even though defining them is one of the contributions of this paper. This planning technique is denoted here by conventional MacEachern.

$$G = \frac{\alpha}{N_T + \alpha} G_0 + \sum_{k=1}^{K_e} \pi_k \delta_{\theta_k} + \sum_{k=K_e+1}^K \pi_k \delta_{\theta_k}. \quad (31)$$

We use the cells coverage and capacity as goodness criteria [7], [8]. As mentioned before, in practice, a cellular

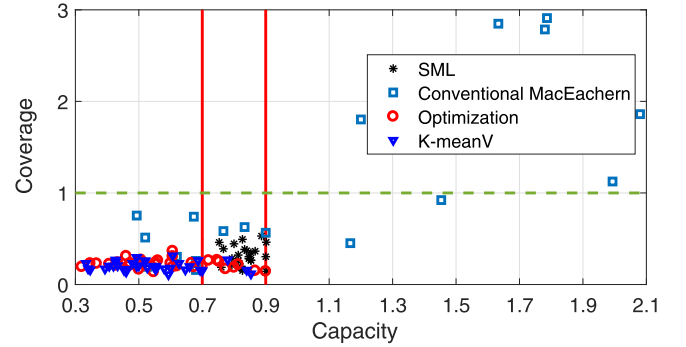


Fig. 4. Dense urban area: coverage and capacity performance.

system is said to be well planned if the network provides a good coverage, i.e., $A_{th} \geq 90\%$ per cell, and good system utilization, i.e., $c_{min} = 70\%$ and $c_{max} = 90\%$ [7], [8]. We set $N_{max} = 150$. This suggests that the number of users per cell has to be in the range $[c_{min}N_{max}, c_{max}N_{max}] = [105, 135]$. The remaining simulation parameters are set as follows. The transmission frequency and bandwidth are set to 2100MHz and 10MHz, respectively. The thermal noise is set to 174dBm. The considered geographical area is assumed to be of size 100Km² in which the users are randomly distributed so that different areas have different densities. P_{max} and γ_{th} are considered to be equal to 10dB and 30dBm, respectively.

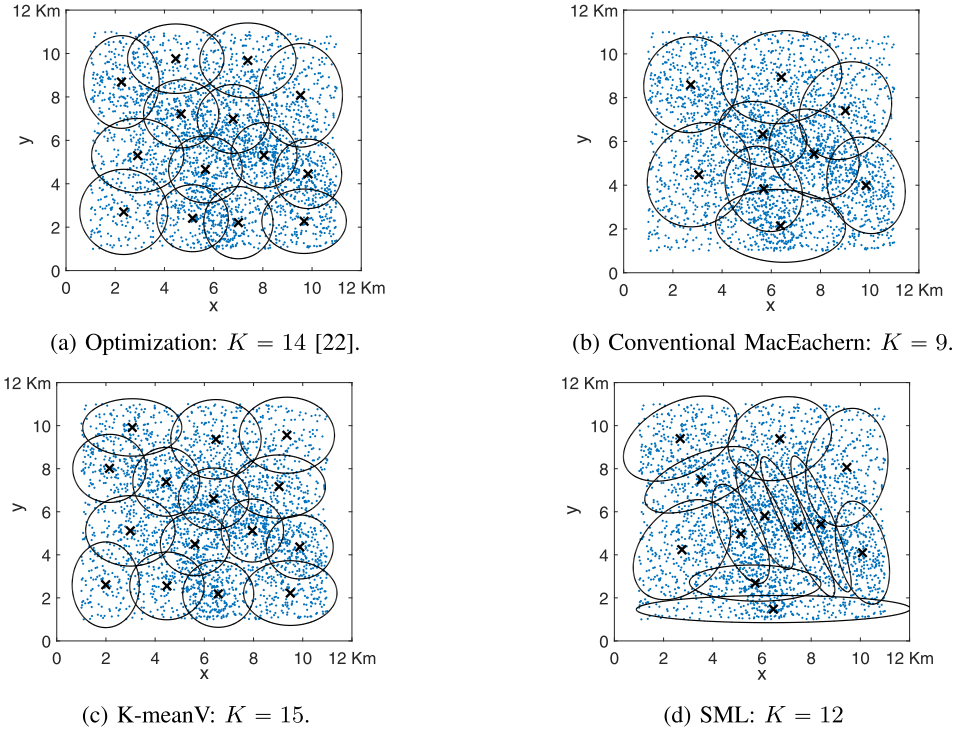


Fig. 5. Urban area: users distribution and association.

A. Proposed Approach for a New Cellular Network Design

We consider here two main scenarios. The first one corresponds to a dense urban area, that is, the areas are capacity limited. The second one corresponds to an urban scenario, where some areas are capacity limited while others are coverage limited.

1) *Dense Urban Area*: We consider $N_T = 3000$ and $N_{\max} = 150$. As such, the [minimum required BSs is $K = \lceil \frac{N_T}{c_{\max} N_{\max}} \rceil = \lceil \frac{3000}{135} \rceil = 23$. For the conventional MacEachern algorithm, we consider the same prior and parameters as the proposed algorithm.

Figs. 3a and 3b show the BSs positions as well as their coverage using the considered optimization and conventional MacEachern techniques, respectively. Figs. 3c and 3d show the same results but for the KmeanV and proposed approach, respectively. From the figures, we can easily conclude that the resources are over provisioned using the optimization and K-meanV techniques as they result in 37 and 34 required BSs, respectively, which are outside the range [23, 29]. Moreover, the conventional MacEachern provides a low number of BSs, that is equal to 19, which is far less than the minimum estimated number, and hence the resources are under provisioned.

We provide in Fig. 4 the performance in terms of the normalized capacity per cell. The number of users per cell is normalized with respect to $N_{\max}$. Recall that we would like to have the normalized number of users per cell to be in the range [0.7, 0.9]. In the figure, the coverage constraint is satisfied if the normalized coverage is less than one, i.e., if it is higher than one it means that a power exceeding the maximum allowed is required to provide the intended coverage. We can

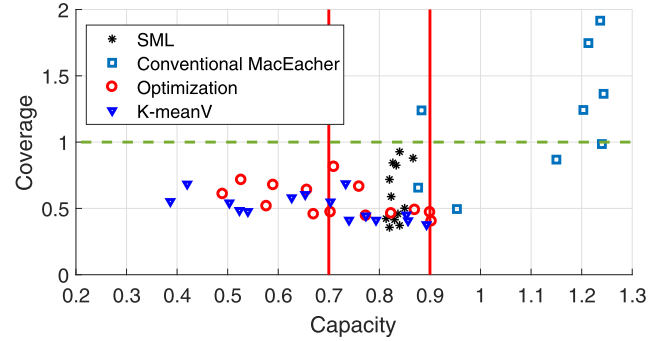


Fig. 6. Urban area: coverage and capacity performance.

observe from the figure that, using K-meanV, the resources are clearly over provisioned, where 34 out of 37 cells have numbers of users less than the minimum value. Moreover, using optimization, $\frac{25}{34} \sim 70\%$ of the cells are over provisioned. It is true that the conventional MacEachern provides the lowest number of required BSs, however, $\frac{16}{19} \sim 86\%$ of the cells are either under or over provisioned. Meanwhile, we can observe that the proposed approach suggests using 24 BSs while satisfying the capacity constraint in all cells. This is a proof of the efficacy of the proposed approach to give a well planned network. Furthermore, it is worth noting that the proposed algorithm also returns the parameters associated to each cell, which are the covariance matrices. This is used to compute the transmission power and the antennas radiation patterns.

2) *Urban Area*: We consider in this section the scenario of an urban area where some sub-areas are capacity limited

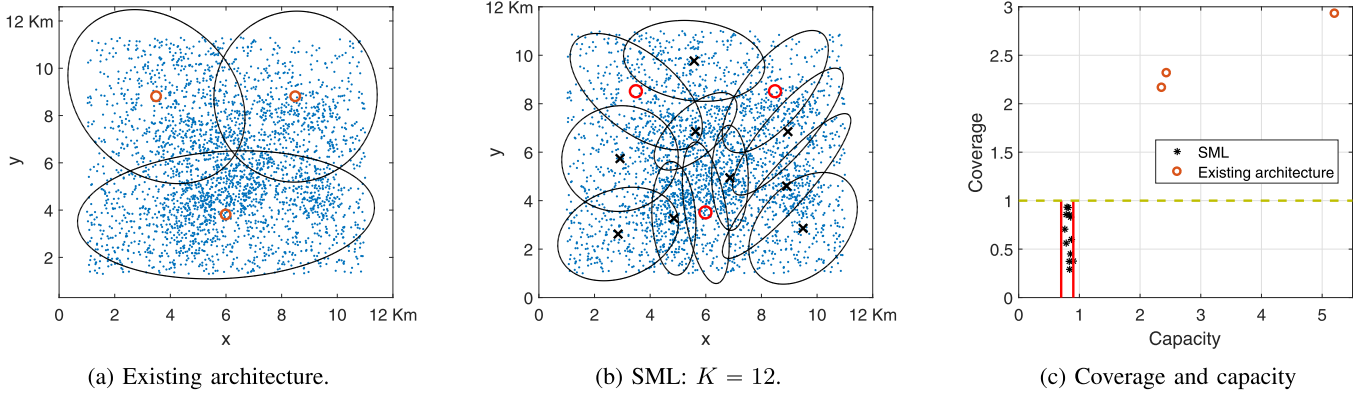


Fig. 7. Supporting an existing architecture.

whereas others are coverage limited. We use $N_T = 1500$, as such, the estimated minimum and maximum of required BSs are respectively $K = \lceil \frac{N_T}{c_{\max} N_{\max}} \rceil = \lceil \frac{1500}{135} \rceil = 12$ and $K = \lceil \frac{N_T}{c_{\min} N_{\min}} \rceil = \lceil \frac{1500}{135} \rceil = 15$. From Fig. 5, with the exception of the conventional MacEachern, all the techniques provide a good estimation of the required number of BSs. This is somewhat expected as, in the urban area, the user distribution is almost uniform and it is easier to plan or at least estimate the range of required number of BSs.

In terms of performance with respect to coverage and capacity, we can see that a few of the cells, using the optimization and K-meanV techniques, are over provisioned (do not respect the capacity criterion.) with ratios $\sim 40\%$ and $\sim 45\%$, respectively, as shown in Figs. 5a and 5c. As for the conventional MacEachern, as shown in Fig. 5b, around 90% of the cells are over provisioned. As for the performance of SML, as shown in Fig. 5d, the size of the cells varies depending on the density of the users. Some cells are large and have coverage up to 92% from its maximum size, whereas some others are of small size with coverage reaching up to 35%.

In Fig. 6, we display the performance of the considered techniques in terms of coverage and capacity. We can see that those constraints are satisfied by all techniques for each cell. However, recall that SML suggests a number of BSs that is less than K-meanV and the considered optimization technique. This demonstrates the efficacy of the proposed framework in utilizing the available resources.

B. Supporting an Existing Architecture

We analyze the performance of the proposed technique where the BSs have to be located to support an existing terrestrial cellular system. We assume that there exists three static BSs located in the following positions $\{(-2.5, 2.5), (0, -2.5), (2.5, 2.5)\}$ Km. We assume that $N_T = 1500$. The simulation results are summarized in Fig. 7. In the figure, the red circles are used to localise the existing BSs, whereas the sign $\times$ is used to localise the new BSs.

Fig. 7 shows that the existing terrestrial system is not able to meet the QoS requirements as the coverage and capacity constraints are not satisfied. Using the proposed approach, the constraints on the capacity and coverage are satisfied

for each cell including the terrestrial ones. This proves the efficacy of the proposed approach for providing a good BSs deployment plan to support an existing terrestrial cellular system.

VIII. CONCLUSION

We proposed in this paper a machine learning-based framework for unsupervised cellular planning. The proposed solution determines the essential planning parameters that are the minimum required BSs, their transmit antenna patterns and the users clusters, while guaranteeing capacity, power and coverage constraints. The proposed solution achieves the intended goal, which is a network setup that gives a good coverage while preventing resources over/under provisioning which is not possible to achieve using existing machine-learning based and optimization based techniques. We have shown that the proposed framework can be used to plan cellular networks from scratch (new network), or expand existing terrestrial networks by adding new BSs to accommodate certain services.

APPENDIX A

For sake of presentation, we consider the case when the center of the ellipse is $(x_{B_k}, y_{B_k}) = (0, 0)$. To obtain the intended results for the general case, one can just follow the steps provided below while replacing (x, y) by $(x_{B_k} - x_{U_i}, y_{B_k} - y_{U_i})$, respectively. The standard deviation of the transmit power of B_k takes the shape of an ellipse described by the following set of points S_k defined in (10).

$$S_k = \left\{ x, y \in \mathbb{R}^2 \mid \frac{1}{1 - (\tau'_k)^2} \left[\frac{x^2}{\zeta_{k,x}^2} + \frac{y^2}{\zeta_{k,y}^2} - \frac{2\tau'_k xy}{\zeta_{k,x}\zeta_{k,y}} \right] = 1 \right\}. \quad (32)$$

Let us denote by $\lambda_{B_k,1}$ and $\lambda_{B_k,2}$ the eigenvalues of Σ_{B_k} . Then, the length of the ellipse axes are $2\sqrt{\lambda_1}$ and $2\sqrt{\lambda_2}$ [36]. An explicit expression of the eigenvalues is given as

$$\lambda_{1,k} = \frac{\zeta_{k,x}^2 + \zeta_{k,y}^2 + \sqrt{(\zeta_{k,x}^2 + \zeta_{k,y}^2)^2 - 4(1 - (\tau'_k)^2)\zeta_{k,x}^2\zeta_{k,y}^2}}{2}, \quad (33)$$

$$\lambda_{2,k} = \frac{\zeta_{k,x}^2 + \zeta_{k,y}^2 - \sqrt{(\zeta_{k,x}^2 + \zeta_{k,y}^2)^2 - 4(1 - (\tau'_k)^2)\zeta_{k,x}^2\zeta_{k,y}^2}}{2}. \quad (34)$$

Let us consider the case when user U_i is situated in position (x_{U_i}, y_{U_i}) . We denote by $p_{B_k, x_{U_i}}^\perp$ and $p_{B_k, x_{U_i}}^\parallel$ the coordinates of $\sqrt{P_{B_k, x_{U_i}}}$ (i.e., standard deviation of the transmit power) on the x and y axes respectively, i.e., $P_{B_k, x_{U_i}} = (p_{B_k, x_{U_i}}^\perp)^2 + (p_{B_k, x_{U_i}}^\parallel)^2$. The tangent of the angle o_i , shown in Fig. 8, can be written as follows.

$$\tan(o_i) = \frac{p_{B_k, x_{U_i}}^\parallel}{p_{B_k, x_{U_i}}^\perp} = \frac{y_{U_i}}{x_{U_i}}. \quad (35)$$

The elements $(p_{B_k, x_{U_i}}^\perp, p_{B_k, x_{U_i}}^\parallel)$ are in $\mathcal{S}_k$, defined in (32). Moreover, they verify the equality in (35). Therefore, we have a system with two unknowns and two independent equalities. Solving this system gives

$$\begin{aligned} (p_{B_k, x_{U_i}}^\parallel)^2 &= \frac{(1 - (\tau'_k)^2)y_{U_i}^2}{\frac{x_{U_i}^2}{\zeta_{k,x}^2} + \frac{y_{U_i}^2}{\zeta_{k,y}^2} - \frac{2\tau'_k x_{U_i} y_{U_i}}{\zeta_{k,x}\zeta_{k,y}}}, \\ (p_{B_k, x_{U_i}}^\perp)^2 &= (p_{B_k, x_{U_i}}^\parallel)^2 \frac{x_{U_i}^2}{y_{U_i}^2} = \frac{(1 - (\tau'_k)^2)x_{U_i}^2}{\frac{x_{U_i}^2}{\zeta_{k,x}^2} + \frac{y_{U_i}^2}{\zeta_{k,y}^2} - \frac{2\tau'_k x_{U_i} y_{U_i}}{\zeta_{k,x}\zeta_{k,y}}}. \end{aligned} \quad (36)$$

Substituting expressions (36) and (36) in $P_{B_k, x_{U_i}} = (p_{B_k, x_{U_i}}^\perp)^2 + (p_{B_k, x_{U_i}}^\parallel)^2$ gives the results in Lemma 1, which concludes the proof.

APPENDIX B

Here, we use ξ'_k to denote $\frac{1}{1 + \frac{\gamma_{th}(\sigma_0^2 + I_k)}{f \times \xi_k}}$. From (6), (13) and the fact that $\Sigma_{B_k} = \xi_k \text{COV}_{B_k}$, the admission rate in the k th cell is provided in (37), as shown at the bottom of this page, where equality (a) comes from the fact that term1 is a Gaussian centred at (x_{B_k}, y_{B_k}) .

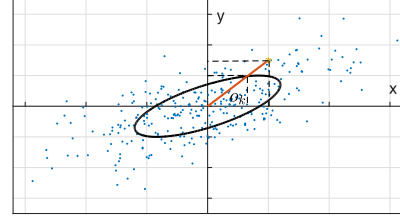


Fig. 8. Transmission power standard deviation.

APPENDIX C

Throughout this proof, we use κ to denote $P_{\max} \left(\frac{f \times (1 - A_{th})}{A_{th} \gamma_{th} (\sigma_0^2 + I_k)} \right)$. The Kullback-Leibler divergence between $\mathcal{N}_\theta$ and $\mathcal{N}_{\theta'}$ is written as

$$\begin{aligned} \mathcal{N}_\theta || \mathcal{N}_{\theta'} &= \int_{x,y \in \mathbb{R}^2} \mathcal{N}_{\theta'}(x,y) \log \left(\frac{\mathcal{N}_{\theta'}(x,y)}{\mathcal{N}_\theta(x,y)} \right) dx dy \\ &= \frac{d}{2} \left[\log \left(\frac{(1 - \tau_k^2) \sigma_{k,x}^2 \sigma_{k,y}^2}{(1 - \tau_k'^2) \sigma_{k,x}'^2 \sigma_{k,y}'^2} \right) - 2 \right. \\ &\quad \left. + \frac{\sigma_{k,x}'^2 \sigma_{k,y}^2 + \sigma_{k,x}^2 \sigma_{k,y}'^2 - 2\tau_k' \tau_k \sigma_{k,x}' \sigma_{k,y}' \sigma_{k,x} \sigma_{k,y}}{(1 - \tau_k'^2) \sigma_{k,x}'^2 \sigma_{k,y}'^2} \right]. \end{aligned} \quad (38)$$

Equality (d) is obtained by invoking the results in [44]. We know that the elements of COV_{B_k} do not respect the condition in (17). This suggests that $(1 - \tau_k^2) \sigma_{k,x}^2 \sigma_{k,y}^2 > \kappa^2$. The goal then is to find the elements of $\mathcal{N}_{\theta'}$ that minimize the distance while verifying (17). To approach $\mathcal{N}_\theta$ as much as possible, it is evident that the elements of the new distribution verify $(1 - \tau_k'^2) \sigma_{k,x}'^2 \sigma_{k,y}'^2 = \kappa^2$. Therefore, minimizing (38) becomes equivalent to minimizing

$$\begin{aligned} \log \left(\frac{(1 - \tau_k^2) \sigma_{k,x}^2 \sigma_{k,y}^2}{\kappa} \right) &+ \frac{\sigma_{k,x}'^2 \sigma_{k,y}^2 + \sigma_{k,x}^2 \sigma_{k,y}'^2 - 2\tau_k' \tau_k \sigma_{k,x}' \sigma_{k,y}' \sigma_{k,x} \sigma_{k,y}}{\kappa} \end{aligned} \quad (39)$$

$$\begin{aligned} A_{B_k} &= \int_{x,y \in \mathbb{R}^2} A_{B_k}(x,y) \mathcal{N}_{\theta_k}(x,y) dx dy \\ &= \int_{x,y \in \mathbb{R}^2} \frac{1}{2\pi \sigma_{k,x} \sigma_{k,y} \sqrt{1 - \tau_k^2}} \exp \left(-\frac{1}{2(1 - \tau_k^2)} \left[\frac{(x_{B_k} - x)^2}{\sigma_{k,x}^2} + \frac{(y_{B_k} - y)^2}{\sigma_{k,y}^2} - \frac{2\tau_k(x_{B_k} - x)(y_{B_k} - y)}{\sigma_{k,x} \sigma_{k,y}} \right] \right) \\ &\quad \times \exp \left(-\frac{\gamma_{th}(\sigma_0^2 + I_k)}{2f \times (1 - (\tau_k)^2)} \left[\frac{(x_{B_k} - x_{U_i})^2}{\xi \sigma_{k,x}^2} + \frac{(y_{B_k} - y_{U_i})^2}{\xi \sigma_{k,y}^2} - \frac{2\tau_k(x_{B_k} - x_{U_i})(y_{B_k} - y_{U_i})}{\xi \sigma_{k,x} \sigma_{k,y}} \right] \right) dx dy \\ &= \xi'_k \int_{x,y \in \mathbb{R}^2} \underbrace{\frac{1}{2\xi'_k \pi \sigma_{k,x} \sigma_{k,y} \sqrt{1 - \tau_k^2}} \exp \left(-\frac{1}{2(1 - \tau_k^2)} \left[\frac{(x_{B_k} - x)^2}{\xi'_k \sigma_{k,x}^2} + \frac{(y_{B_k} - y)^2}{\xi'_k \sigma_{k,y}^2} - \frac{2\tau_k(x_{B_k} - x)(y_{B_k} - y)}{\xi'_k \sigma_{k,x} \sigma_{k,y}} \right] \right)}_{\text{term1}} \\ &\stackrel{a}{=} \xi'_k = \frac{1}{1 + \frac{\gamma_{th}(\sigma_0^2 + I_k)}{f \times \xi_k}}. \end{aligned} \quad (37)$$

Since the first term in (39) is a constant, we need to just minimize over the second term while removing κ in the denominator. Moreover, given the equality $(1 - \tau_k'^2)\sigma_{k,x}'^2\sigma_{k,y}'^2 = \kappa^2$, one can replace $\sigma_{k,x}'\sigma_{k,y}'$ by $\frac{\kappa}{\sqrt{1-\tau_k'^2}}$. Using traditional mathematical operations, minimizing over the second term is equivalent to minimizing

$$(\sigma_{k,x}'\sigma_{k,y}' - \sigma_{k,x}'\sigma_{k,x})^2 + \frac{2(1 - \tau_k'\tau_k)\sigma_{k,y}\sigma_{k,x}\kappa}{\sqrt{1 - \tau_k'^2}}. \quad (40)$$

In (40), the two terms are positive and contain completely independent variables to minimize over. In fact, in the first term, we have $\sigma_{k,x}'$ and $\sigma_{k,y}'$, whereas in the second term we have only τ_k' . Consequently, minimizing (40) is equivalent to minimizing the first term and the second term separately. Minimizing the first term gives $\frac{\sigma_{k,x}'}{\sigma_{k,y}'} = \frac{\sigma_{k,x}}{\sigma_{k,y}}$. This suggests that there exists a positive constant μ' such that $\sigma_{k,x}' = \mu'\sigma_{k,x}$ and $\sigma_{k,y}' = \mu'\sigma_{k,y}$. For the second term, it is sufficient to provide the element that minimizes $\frac{1-\tau_k'\tau_k}{\sqrt{1-\tau_k'^2}}$. Setting the first derivative to zero and analyzing the second derivative, we find that the minimum is achieved when $\tau_k' = \tau_k$. Collecting both findings gives the following set of equations.

$$\begin{cases} \sigma_{k,x}' = \mu'\sigma_{k,x}, & \sigma_{k,y}' = \mu'\sigma_{k,y} \\ \tau_k' = \tau_k, & (1 - \tau_k'^2)\sigma_{k,x}'^2\sigma_{k,y}'^2 = \kappa^2. \end{cases} \quad (41)$$

Substituting the first three equalities in the fourth one gives

$$(1 - \tau_k^2)(\mu')^2\sigma_{k,x}^2\sigma_{k,y}^2 = \kappa^2 \Rightarrow \mu' = \frac{\kappa}{\sqrt{1 - \tau_k^2}\sigma_{k,x}\sigma_{k,y}} = \mu. \quad (42)$$

This completes the proof.

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